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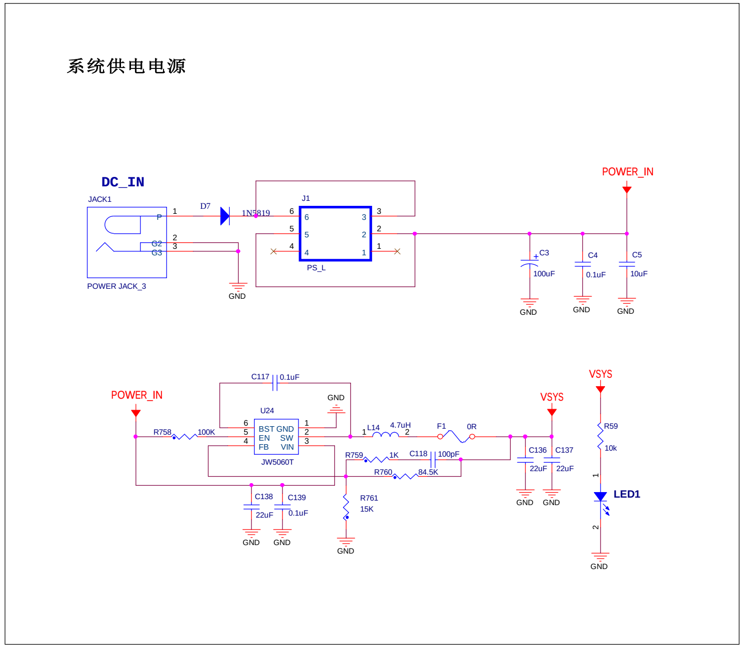
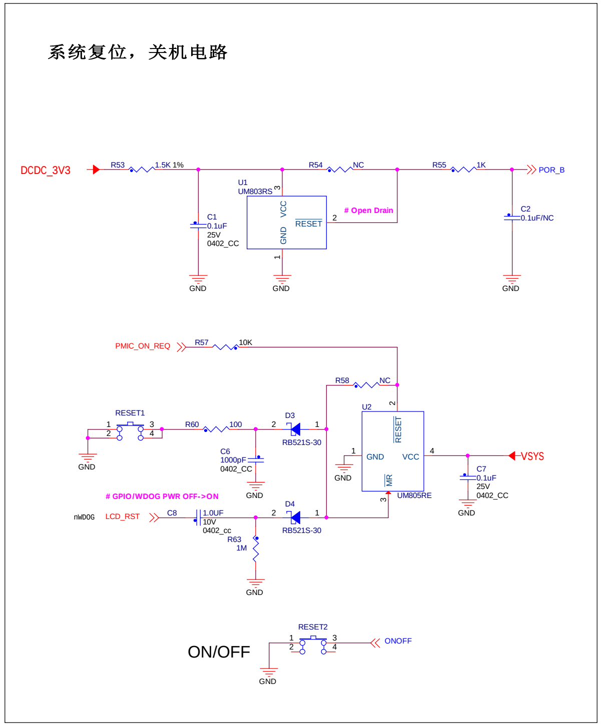
Revision History

Rev. Code	*Date*	*Description*
V1.0	*2020-03-23*	*Revision 1.0 release*

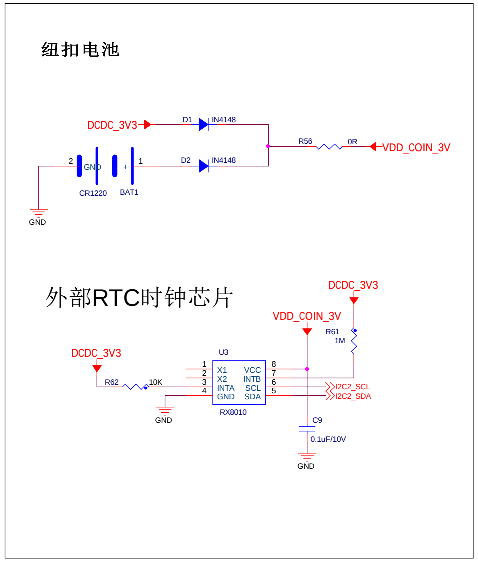
系统供电电源



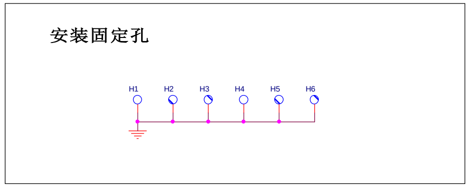
系统复位，关机电路



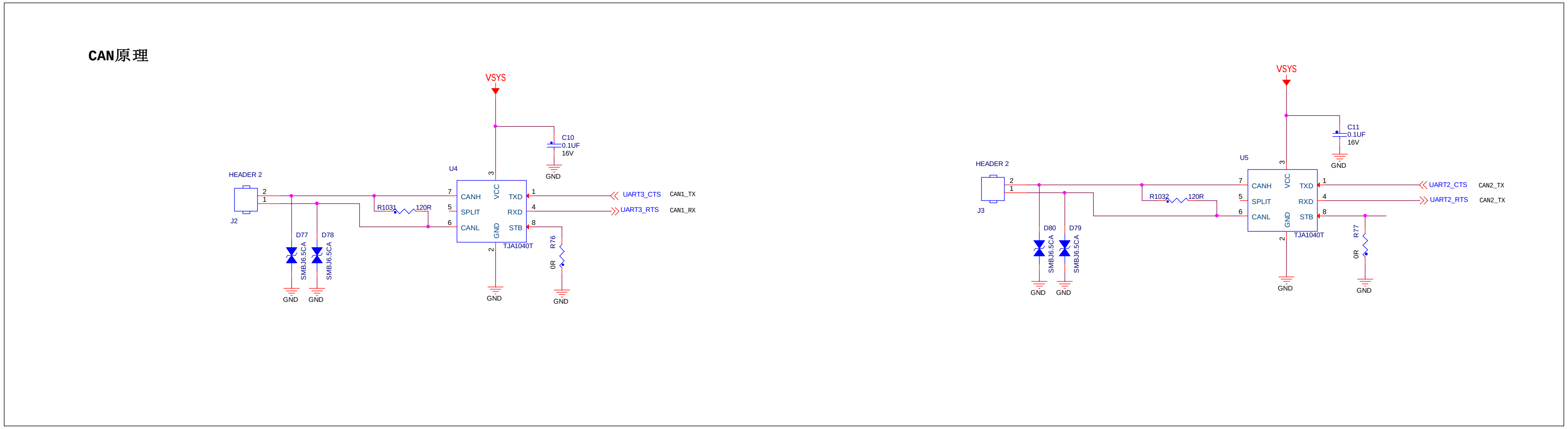
纽扣电池



安装固定孔



CAN原理



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Title	DC_JACK CAN		
Size A2	Document Number		Rev 1.0
Date: Saturday, March 28, 2020	Sheet 2 of 14		

FUSE MAP

<Default: QSPI BOOT>

	0/1	0/1	0/1	1	0	0	0	0
TYPE	**BOOT_CFG1[7]**	**BOOT_CFG1[6]**	**BOOT_CFG1[5]**	**BOOT_CFG1[4]**	**BOOT_CFG1[3]**	**BOOT_CFG1[2]**	**BOOT_CFG1[1]**	**BOOT_CFG1[0]**
QSPI	0	0	0	1	Reserved	DDRSMP: "000" : Default "001-111"		
WEIM	0	0	0	0	Memory Type: 0 - NOR Flash 1 - OneNAND	Reserved	Reserved	Reserved
Serial-ROM	0	0	1	1	Reserved	Reserved	Reserved	Reserved
SD/eSD	0	1	0	Fast Boot: 0 - Regular 1 - Fast Boot	SD/SDXC Speed 00 - Normal/SDR12 01 - High/SDR25 10 - SDR50 11 - SDR104		SD Power Cycle Enable '0' - No power cycle '1' - Enabled via USDHC_RST pad (uSDHC3 & 4 only)	SD Loopback Clock Source Sel(for SDR50 and SDR104 only) '0' - through SD pad '1' - direct
MMC/eMMC	0	1	1	Fast Boot: 0 - Regular 1 - Fast Boot	SD/MMC Speed 0 - Highl 1- Normal	Fast Boot Acknowledge Disable: 0 - Boot Ack Enabled 1 - Boot Ack Disabled	SD Power Cycle Enable '0' - No power cycle '1' - Enabled via USDHC_RST pad (uSDHC3 & 4 only)	SD Loopback Clock Source Sel(for SDR50 and SDR104 only) '0' - through SD pad '1' - direct
NAND	1	BT_TOGGLEMODE	Pages In Block: 00 - 128 01 - 64 10 - 32 11 - 256		Nand Number Of Devices: 00 - 1 01 - 2 10 - 4 11 - Reserved		Nand_Row_address_bytes: 00 - 3 01 - 2 10 - 4 11 - 5	

	0	0	0	0	1	0	0	0
TYPE	**BOOT_CFG2[7]**	**BOOT_CFG2[6]**	**BOOT_CFG2[5]**	**BOOT_CFG2[4]**	**BOOT_CFG2[3]**	**BOOT_CFG2[2]**	**BOOT_CFG2[1]**	**BOOT_CFG2[0]**
QSPI	Reserved	HSPHS: Half Speed Phase Selection 0 : select sampling at non-inverted clock 1: select sampling at inverted clock	HSDLY: Half Speed Delay selection 0 : one clock delay 1: two clock delay	FSPHS: Full Speed Phase Selection 0 : select sampling at non-inverted clock 1: select sampling at inverted clock	FSDLY: Full Speed Delay selection 0: one clock delay 1: two clock delay	Boot Frequencies (ARM/DDR) 0 - 500 / 400 MHz 1 - 250 / 200 MHz	Reserved	Reserved
WEIM	Muxing Scheme: 00 - A/D16 01 - A+DH 10 - A+DL 11- Reserved		OneNand Page Size: 00 - 1KB 01 - 2KB 10 - 4KB 11 - Reserved		Reserved	Boot Frequencies (ARM/DDR) 0 - 500 / 400 MHz 1 - 250 / 200 MHz	Reserved	Reserved
Serial-ROM	Reserved	Reserved	Reserved	Reserved	Reserved	Boot Frequencies (ARM/DDR) 0 - 500 / 400 MHz 1 - 250 / 200 MHz	Reserved	Reserved
SD/eSD	SD Calibration Step '00' - 1 TBD		Bus Width: 0 - 1-bit 1 - 4-bit	Port Select: 00 - eSDHC1 01 - eSDHC2 10 - Reserved 11 - Reserved		Boot Frequencies (ARM/DDR) 0 - 500 / 400 MHz 1 - 250 / 200 MHz	SD1 VOLTAGE SELECTION 0 - 3.3V 1 - 1.8V	Reserved
MMC/eMMC	Bus Width: 000 - 1-bit 001 - 4-bit 010 - 8-bit 101 - 4-bit DDR (MMC 4.4) 110 - 8-bit DDR (MMC 4.4) Else - reserved.			Port Select: 00 - eSDHC1 01 - eSDHC2 10 - Reserved 11 - Reserved		Boot Frequencies (ARM/DDR) 0 - 500 / 400 MHz 1 - 250 / 200 MHz	SD1 VOLTAGE SELECTION 0 - 3.3V 1 - 1.8V	Reserved
NAND	Toggle Mode 33MHz Preamble Delay, Read Latency: '000' - 16 GPMICLK cycles. '001' - 1 GPMICLK cycles. '010' - 2 GPMICLK cycles. '011' - 3 GPMICLK cycles. '100' - 4 GPMICLK cycles. '101' - 5 GPMICLK cycles. '110' - 6 GPMICLK cycles. '111' - 7 GPMICLK cycles.			BOOT_SEARCH_COUNT: 00 - 2 01 - 2 10 - 4 11 - 8		Boot Frequencies (ARM/DDR) 0 - 500 / 400 MHz 1 - 250 / 200 MHz	Reset Time '0' - 12ms '1' - 22ms (LBA Nand)	Reserved

	0	0	0	0	0	0	0	0
TYPE	**BOOT_CFG4[7]**	**BOOT_CFG4[6]**	**BOOT_CFG4[5]**	**BOOT_CFG4[4]**	**BOOT_CFG4[3]**	**BOOT_CFG4[2]**	**BOOT_CFG4[1]**	**BOOT_CFG4[0]**
0x450	Infinit-Loop (Debug USE only) 0 - Disable 1- Enable	EEPROM Recovery Enable '0' - Disabled '1' - Enabled	CS select (SPI only): 00 - CS#0 (default) 01 - CS#1 10 - CS#2 11 - CS#3		SPI Addressing: 0 - 2-bytes (16-bit) 1 - 3-bytes (24-bit)	Port Select: 000 - eCSPI1 001 - eCSPI2 010 - eCSPI3 011 - eCSPI4 100 - Reserved 101 - Reserved 110 - Reserved 111 - Reserved		
0x460	L2_HW_INVALIDATE_DISABLE	Reserved	FORCE_COLD_BOOT (Reflected in SBMR2)	BT_FUSE_SEL	DIR_BT_DIS	Reserved	SEC_CONFIG[1]	Reserved
0x460	Reserved (DDR3 config options)							
0x460	JTAG_SMODE[1:0]	WDOG_ENABLE '0' - Disabled '1' - Enabled	SJC_DISABLE	Reserved	Reserved	Reserved	Reserved	Reserved
0x460	Reserved	Reserved	Reserved	TZASC_ENABLE	JTAG_HEO	KTE	Reserved	DLL_ENABLE 0 - Disable DLL for SD/eMMC 1 - Enable DLL for SD/Emmc
0x470	DLL Override: 0 - DLL Slave Mode for SD/eMMC 1 - DLL Override Mode for SD/eMMC	Reserved	SD2 VOLTAGE SELECTION 0 - 3.3V 1 - 1.8V	Reserved	Disable SDMMC Manufacture mode 0 - Enable 1 - Disable	L1 I-Cache DISABLE	BT_MMU_DISABLE	Override Pad Settings (using PAD_SETTINGS value)
0x470	Reserved for unexpected requirements	eMMC 4.4 - RESET TO PRE-IDLE STATE	Override HYS bit for SD/MMC pads	USDHC_PAD_PULL_DOWN 0 - no action 1 - pull down	ENABLE_EMMC_22K_PULLUP 0 - 47K pullup 1 - 22K pullup	ADD_DS_SET_GPR1_16 0 - Set 1 - Don't set	USDHC_IOMUX_SION_BIT_ENABLE 0 - Disable 1 - Enable	USDHC IOMUX SRE Enable 0 - Disable 1 - Enable
0x470	USDHC_CMD_OE_PRE_EN (SD/MMC debug)	LPB_BOOT (Core / DDR- Bus) '00' - LPB Disable '01' - 1 GPIO (def freq) '10' - Div by2 '11' - Div by 4		BT_LPB_POLARITY (GPIO polarity)	POWER_MNG_CFG (LDO's DCDC's) (Reserved - NOT USED)			
0x470	Override NAND Pad Settings (using PAD_SETTINGS value)	MMC_DLL_DLY[6:0] Delay target for SD/eMMC DLL, it is applied to slave mode target delay or override mode target delay depends on DLL Override fuse bit value.						

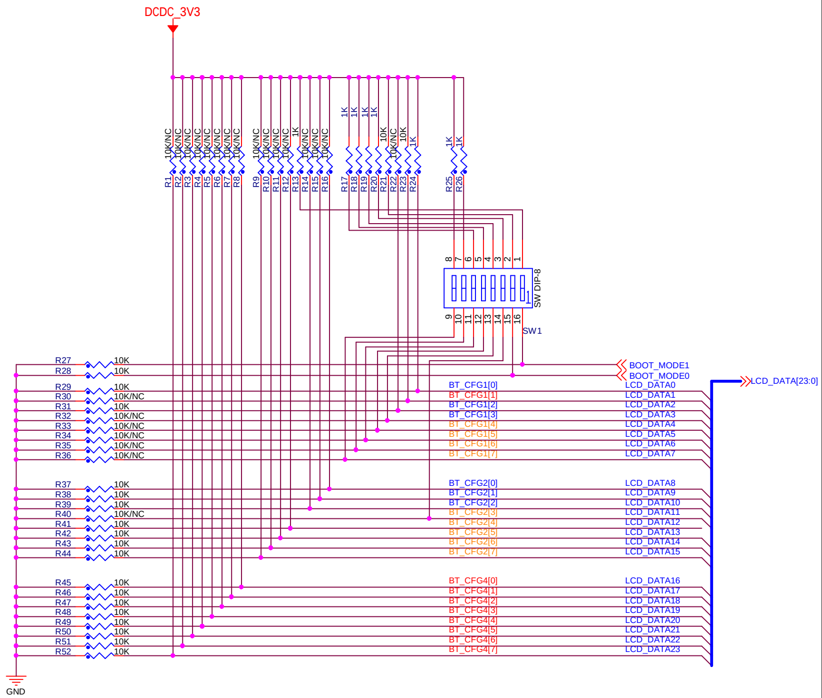
启动方式

NAND MT29F32G08CBACA

1 page = (4K + 224 bytes)
1 block = (4K + 224) bytes x 256 pages
= (1024K + 56K) bytes
1 plane = (1024K + 56K) bytes x 2048 blocks
= 17,280Mb
1 LUN = 17,280Mb x 2 planes
= 34,560Mb

BMODE[1:0]	BOOT TYPE
00	Boot From Fuses
01	Serial Downloader
10	Internal Boot (Development)
11	Reserved

Boot Configuration



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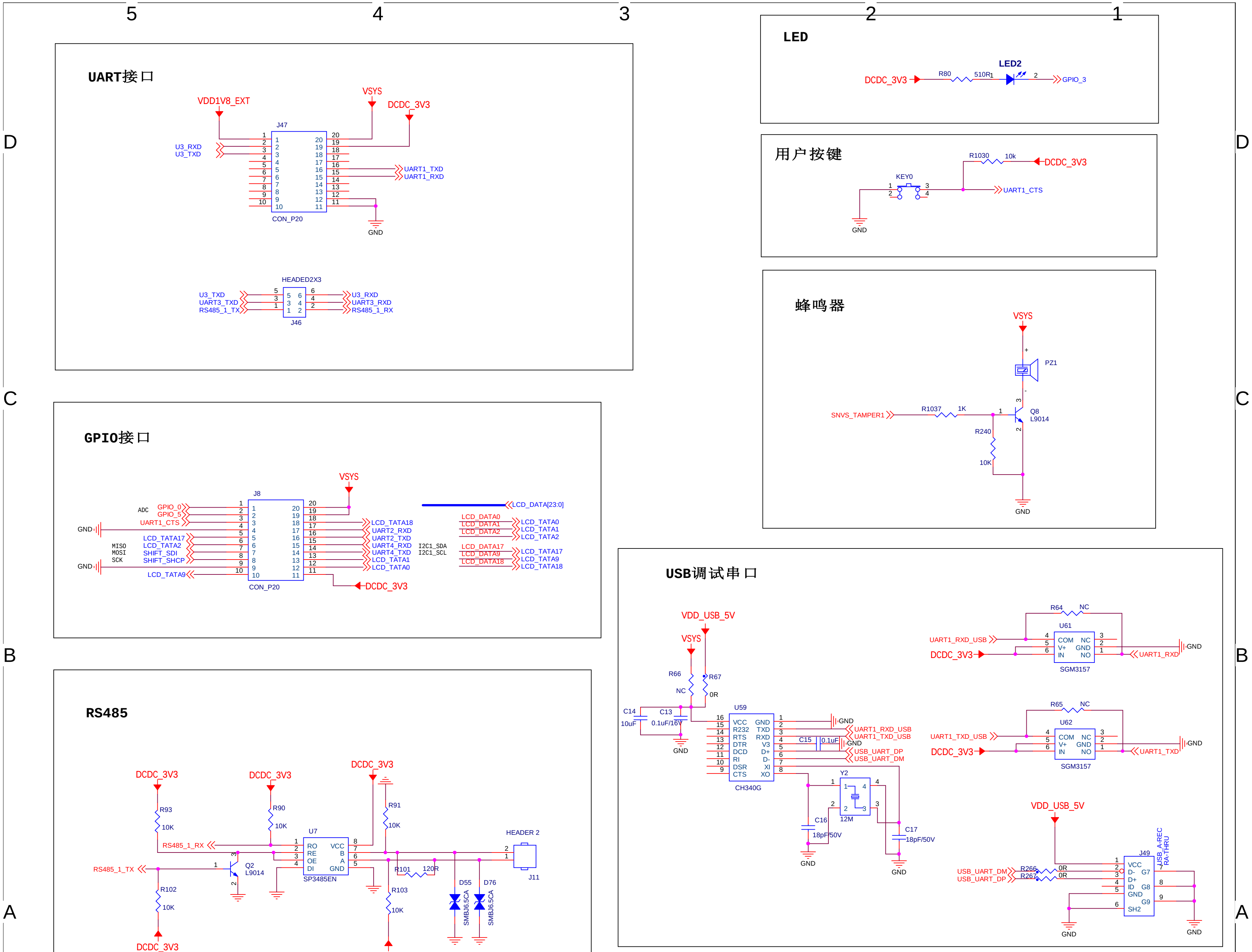
Title	BOOT CFG	
Size C	Document Number	Rev 1.0
Date: Tuesday, March 24, 2020	Sheet 3 of 14	



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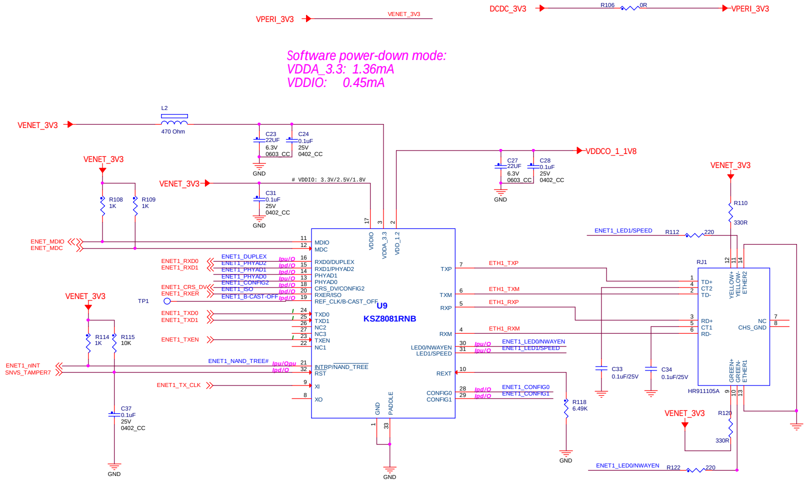
Title: UARTs /GPIO

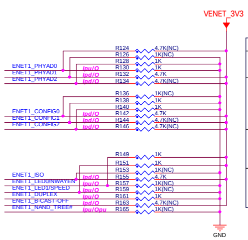
Size	Document Number	Rev
Custom		1.0

Date: Tuesday, March 24, 2020 Sheet 4 of 14

100M ETHERNET RMII PHY x1



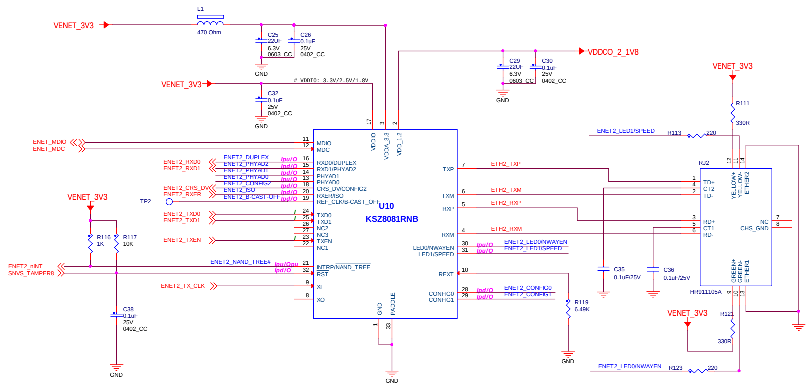
RMII CFG



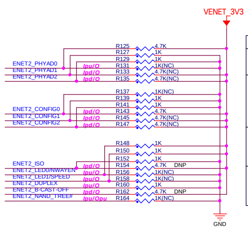
# CFG	Description
PHYAD[2:0]	PHY ADDR 00-XXX (00001 DEFAULT)
CONFIG[2:0]	IF MODE 001 RMII 101 RMII Back-to-Back xxx Reserved-not used
ISO	ISOLATE mode Pull-up = Enable Pull-down (defaule) = Disable
SPEED	SPEED mode Pull-up (defaule) = 100Mbps Pull-down = 10Mbps

# CFG	Description
DUPLEX	DUPLEX mode Pull-up (defaule) = Half Duplex Pull-down = Full Duplex
NWAYEN	Nway Auto-Negotiation Pull-up (defaule) = Enable Pull-down = Disable
B_CAST_OFF	Broadcast Off - for PHY Address 0 Pull-up = PHY Address 0 set as unique PHY addr Pull-down (default) = PHY Address 0 set as broadcast PHY addr
NAND_TREE#	NAND Tree Mode Pull-up (defaule) = Disable Pull-down = Enable

100M ETHERNET RMII PHY x2



RMII CFG



# CFG	Description
PHYAD[2:0]	PHY ADDR 00-XXX (00001 DEFAULT)
CONFIG[2:0]	IF MODE 001 RMII 101 RMII Back-to-Back xxx Reserved-not used
ISO	ISOLATE mode Pull-up = Enable Pull-down (defaule) = Disable
SPEED	SPEED mode Pull-up (defaule) = 100Mbps Pull-down = 10Mbps

# CFG	Description
DUPLEX	DUPLEX mode Pull-up (defaule) = Half Duplex Pull-down = Full Duplex
NWAYEN	Nway Auto-Negotiation Pull-up (defaule) = Enable Pull-down = Disable
B_CAST_OFF	Broadcast Off - for PHY Address 0 Pull-up = PHY Address 0 set as unique PHY addr Pull-down (default) = PHY Address 0 set as broadcast PHY addr
NAND_TREE#	NAND Tree Mode Pull-up (defaule) = Disable Pull-down = Enable

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Title: ETHERNET

Size: D | Document Number: | Rev: 1.0

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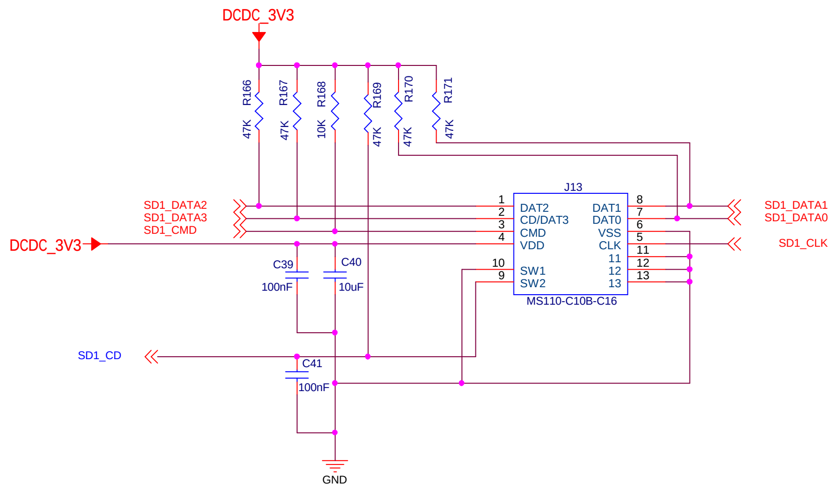
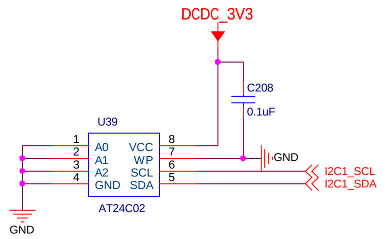
TF Card



EEPROM



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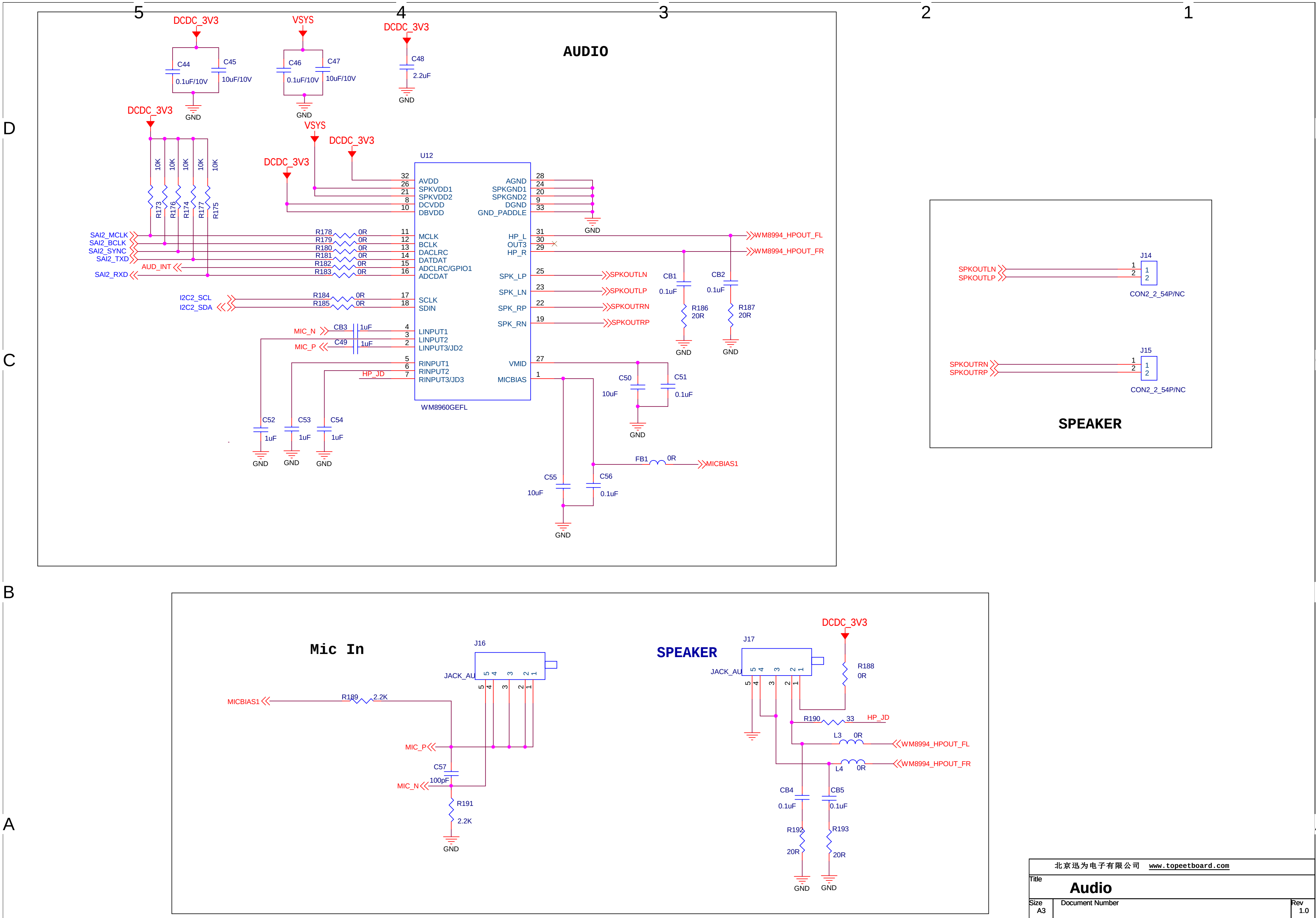
Title	TF Card, EEPROM	
Size A3	Document Number	Rev 1.0
Date: Tuesday, March 24, 2020	Sheet 6 of 14	



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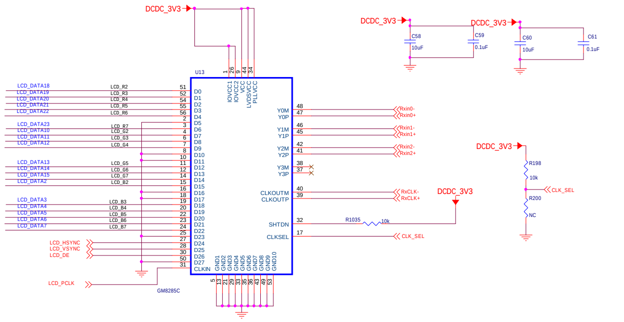
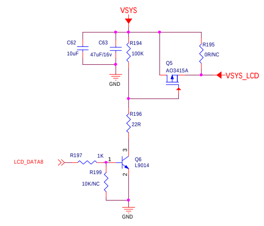
Title	Audio		
Size A3	Document Number		Rev 1.0
Date: Tuesday, March 24, 2020	Sheet 7	of 134	

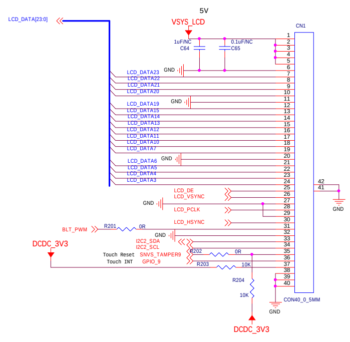
RGB TO LVDS



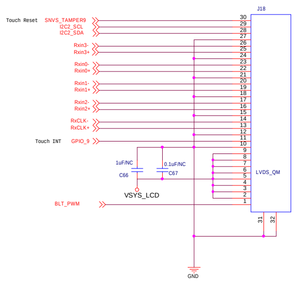
屏幕电源控制



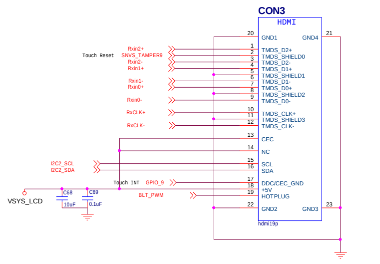
RGB接口



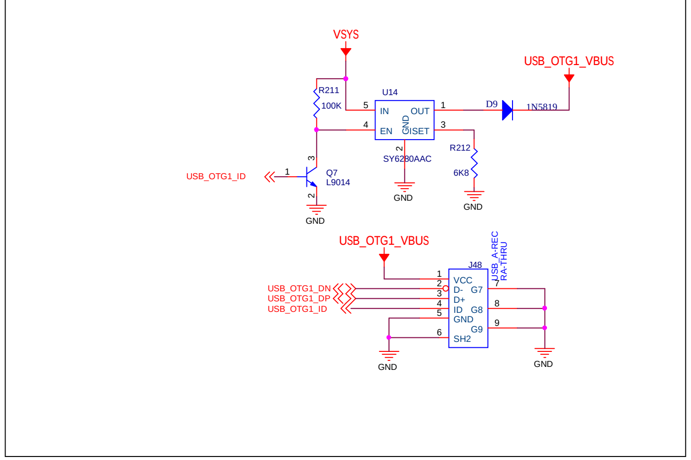
LVDS FPC接口方式



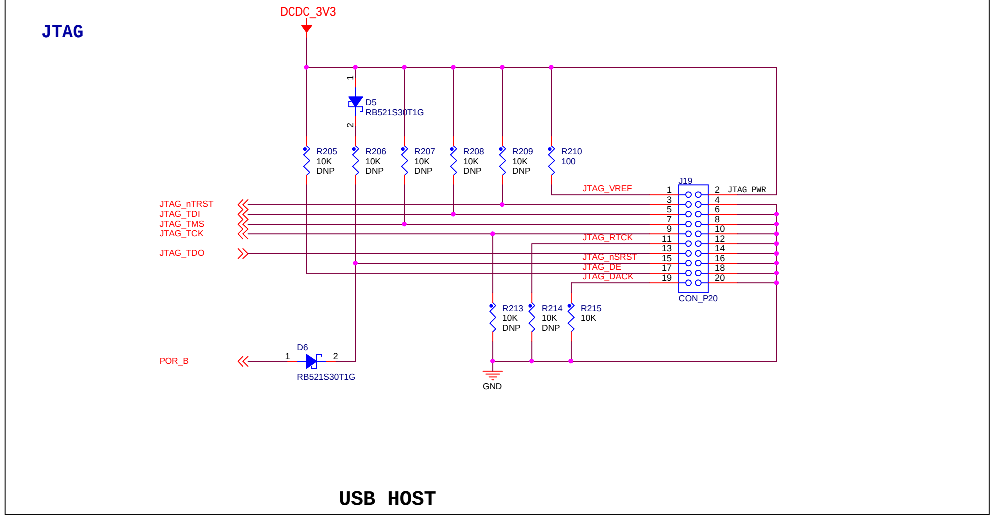
LVDS HDMI接口方式



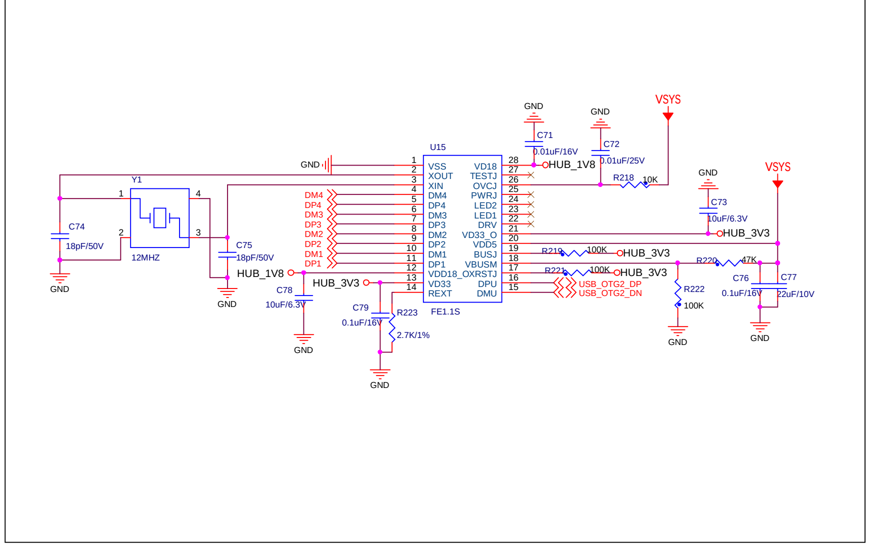
USB OTG



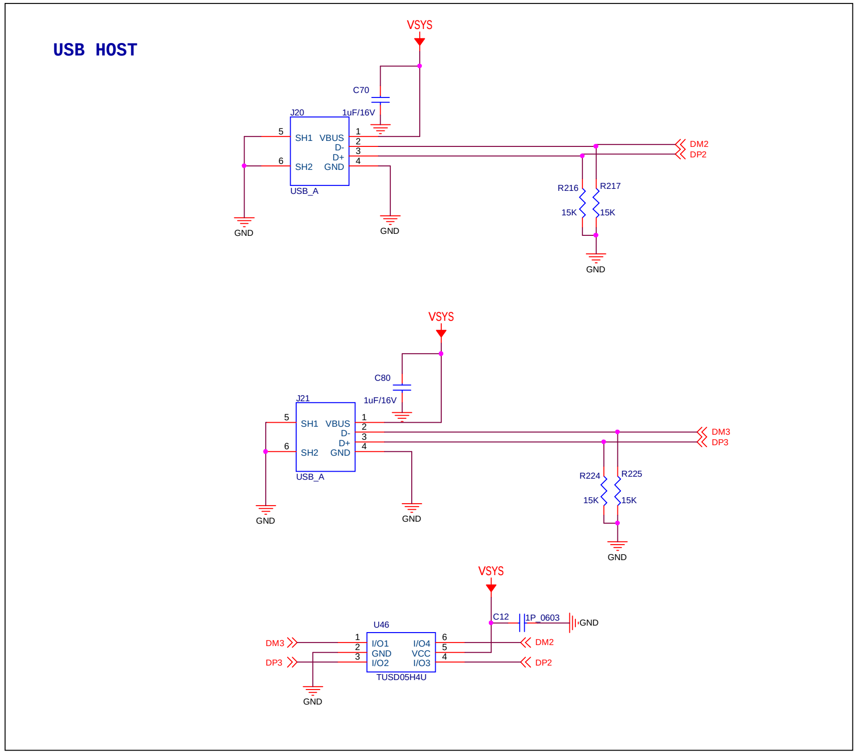
JTAG



USB HUB



USB HOST



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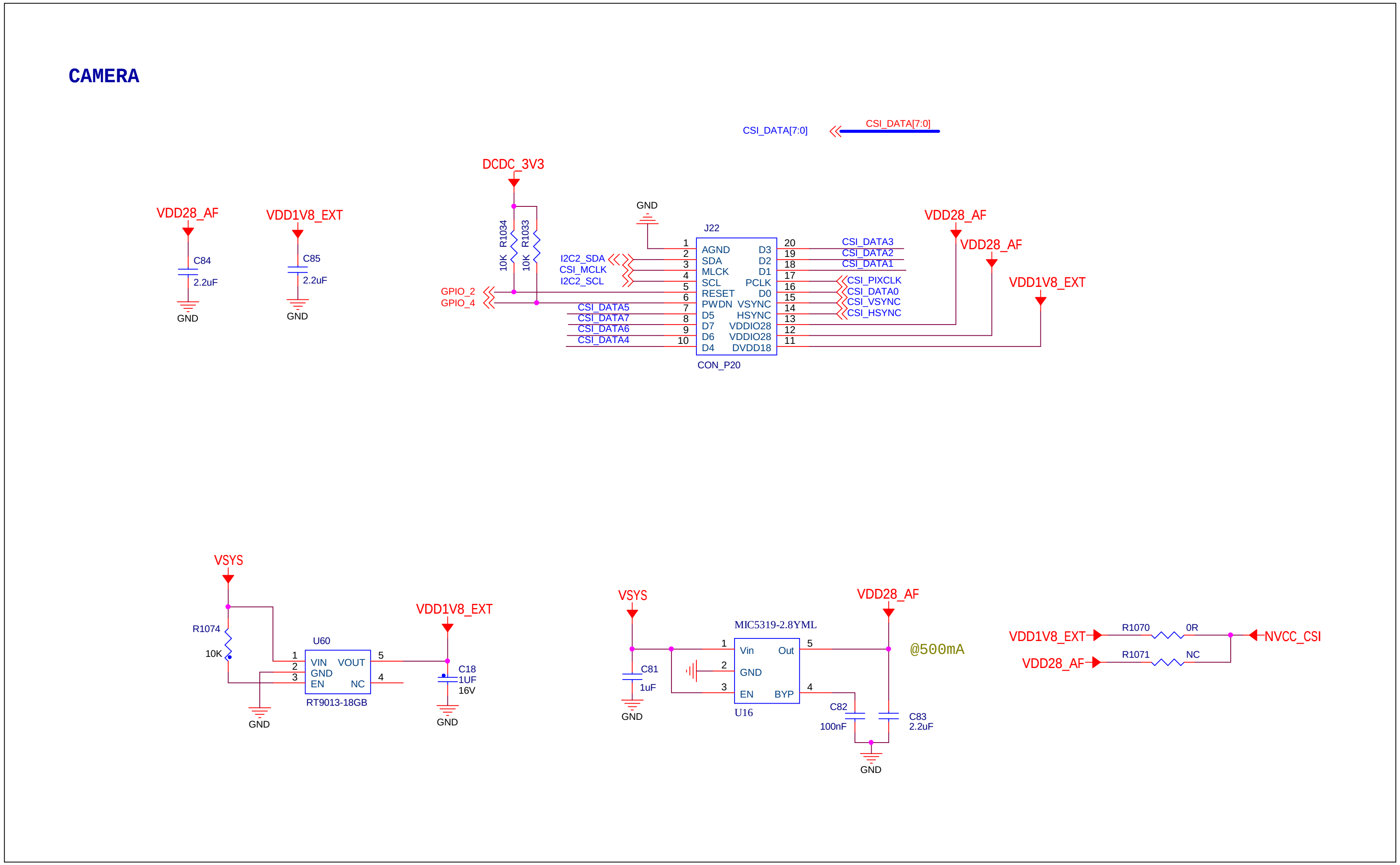
Title	USB_JTAG	
Size C	Document Number	Rev 1.0
Date: Tuesday, March 24, 2020	Sheet 9 of 14	



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Title: CAMERA

Size: A3 | Document Number | Rev: 1.0

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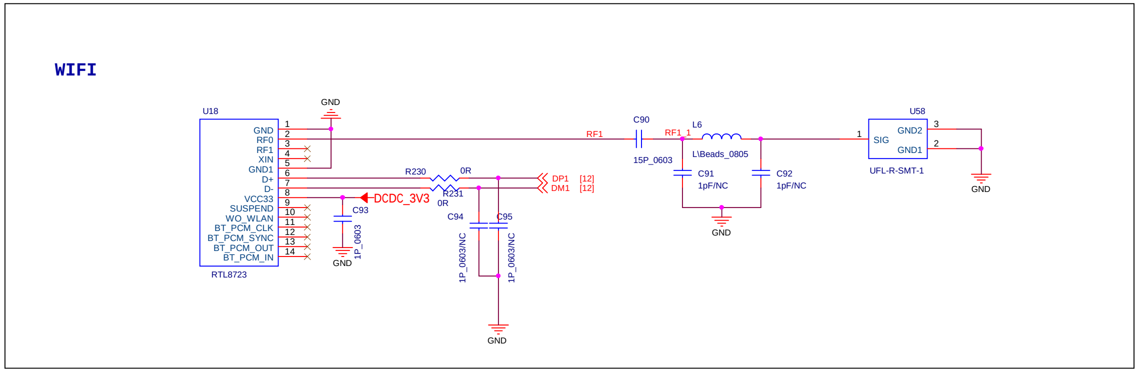

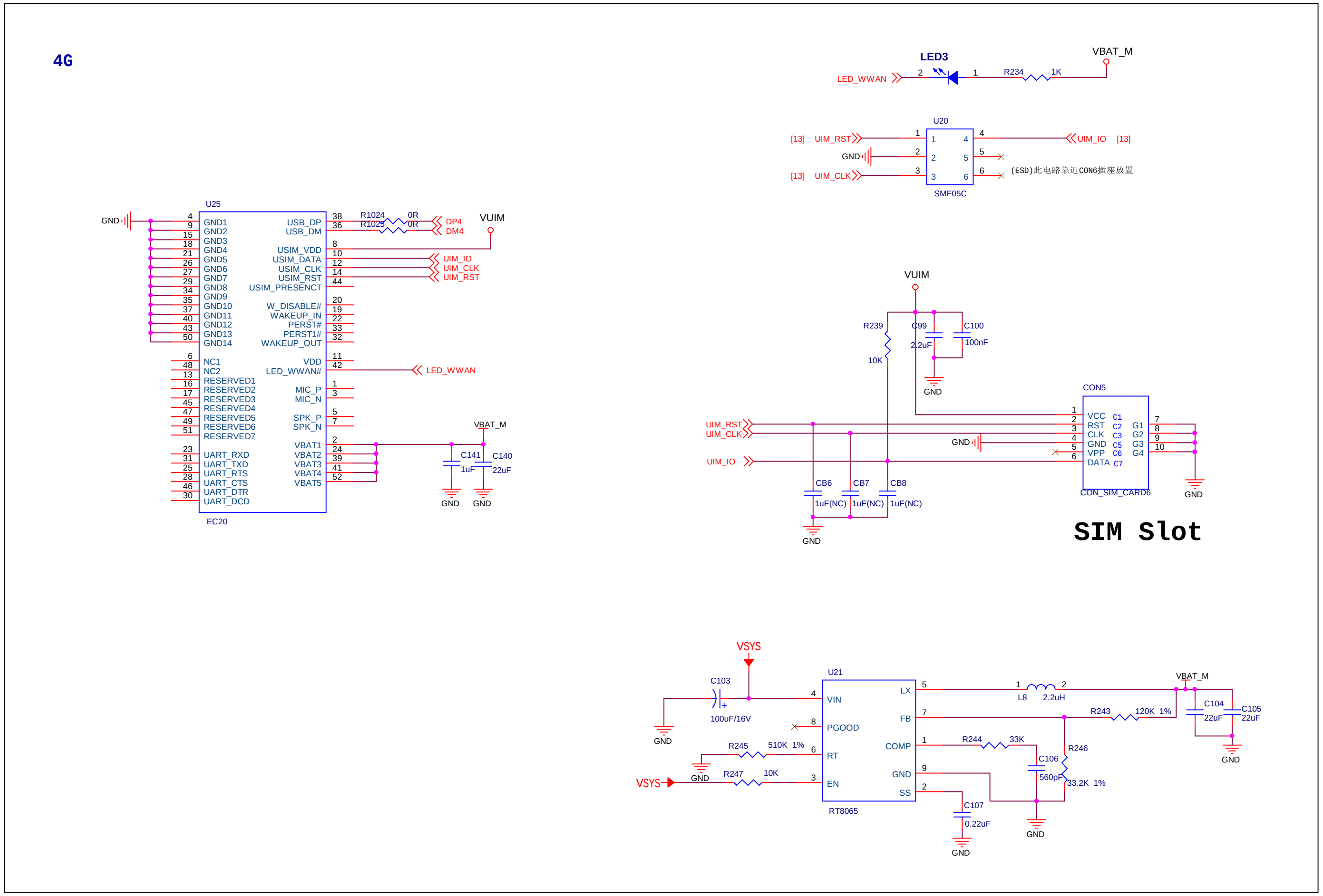

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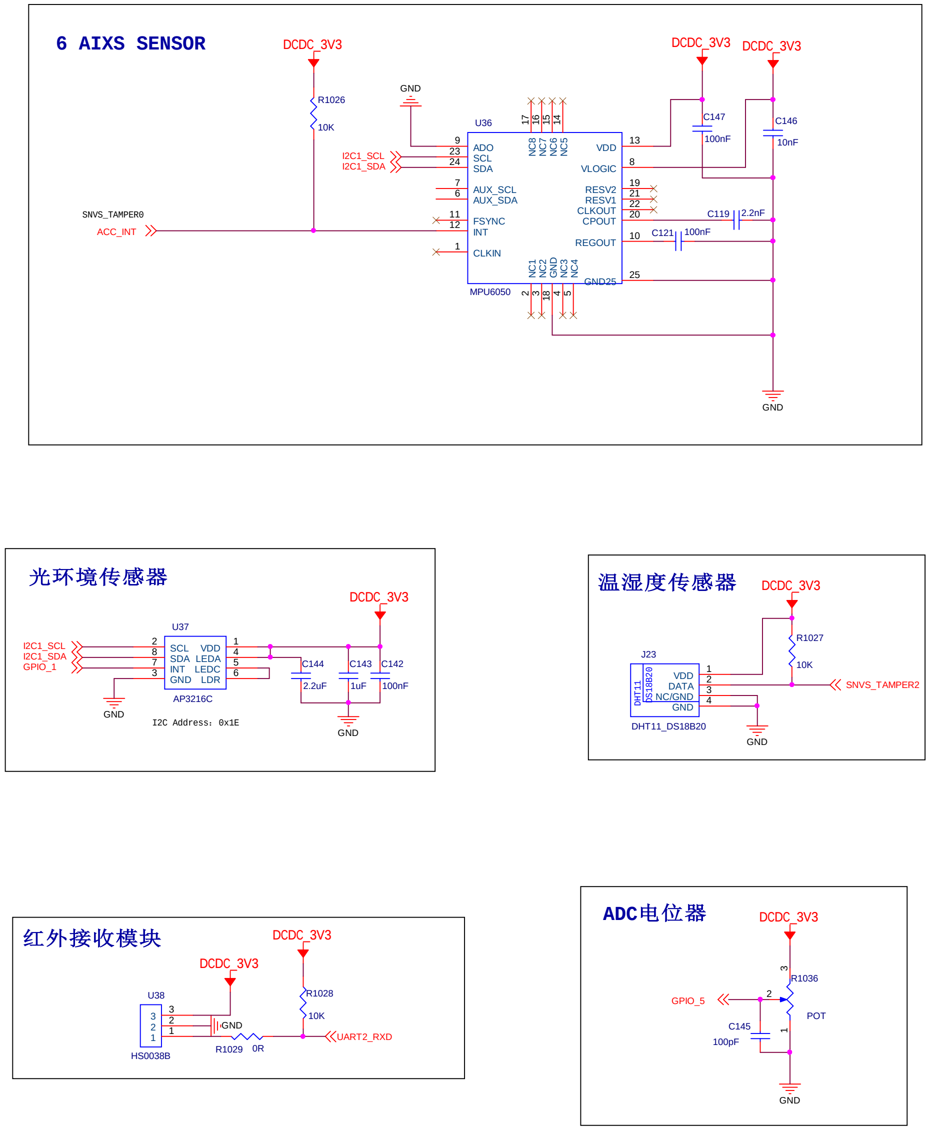
Title: WIFI/4G

Size	Document Number	Rev
C	<Doc>	1.0

Date: Tuesday, March 24, 2020 | Sheet 11 of 14



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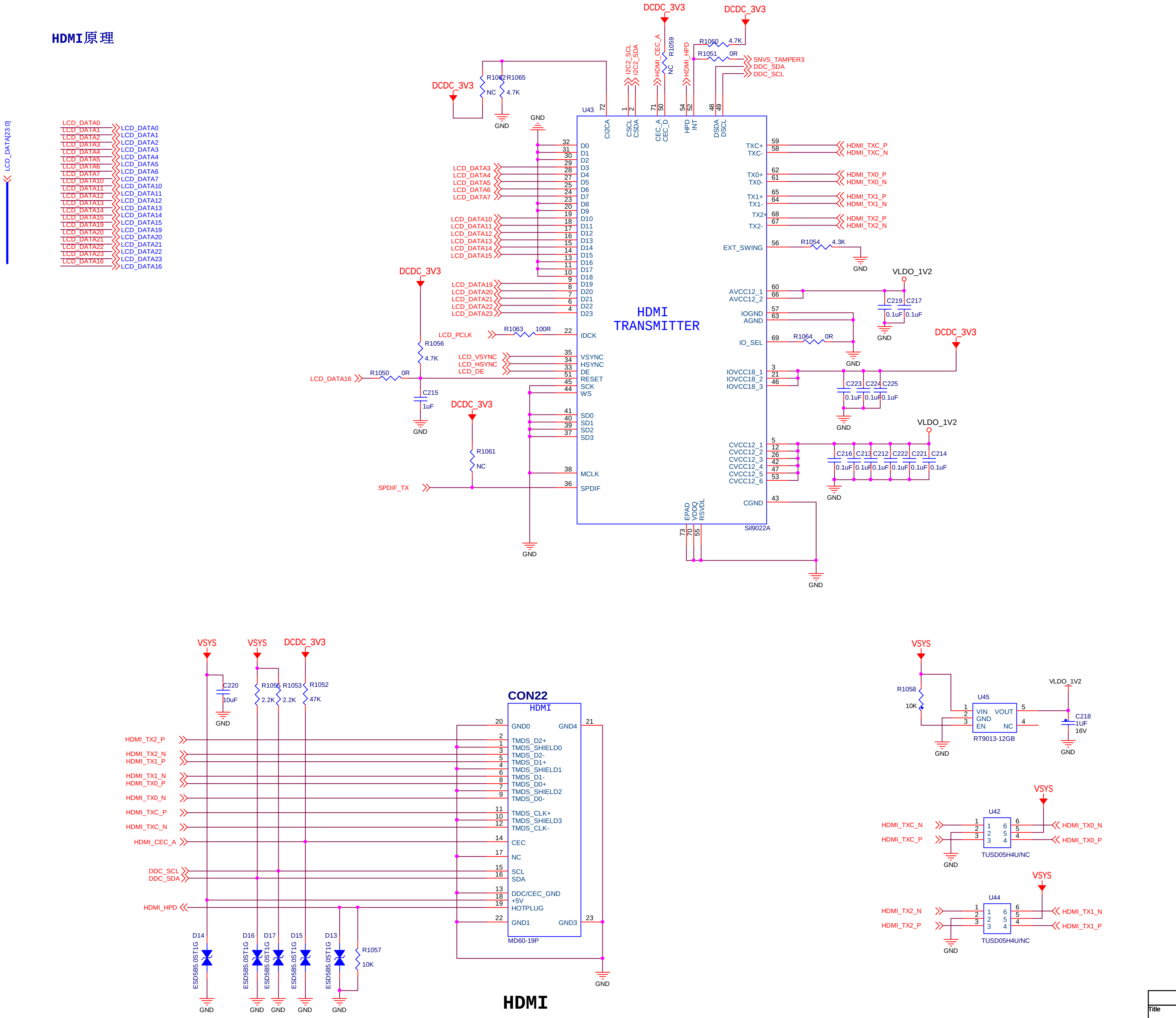
Title: SENSOR

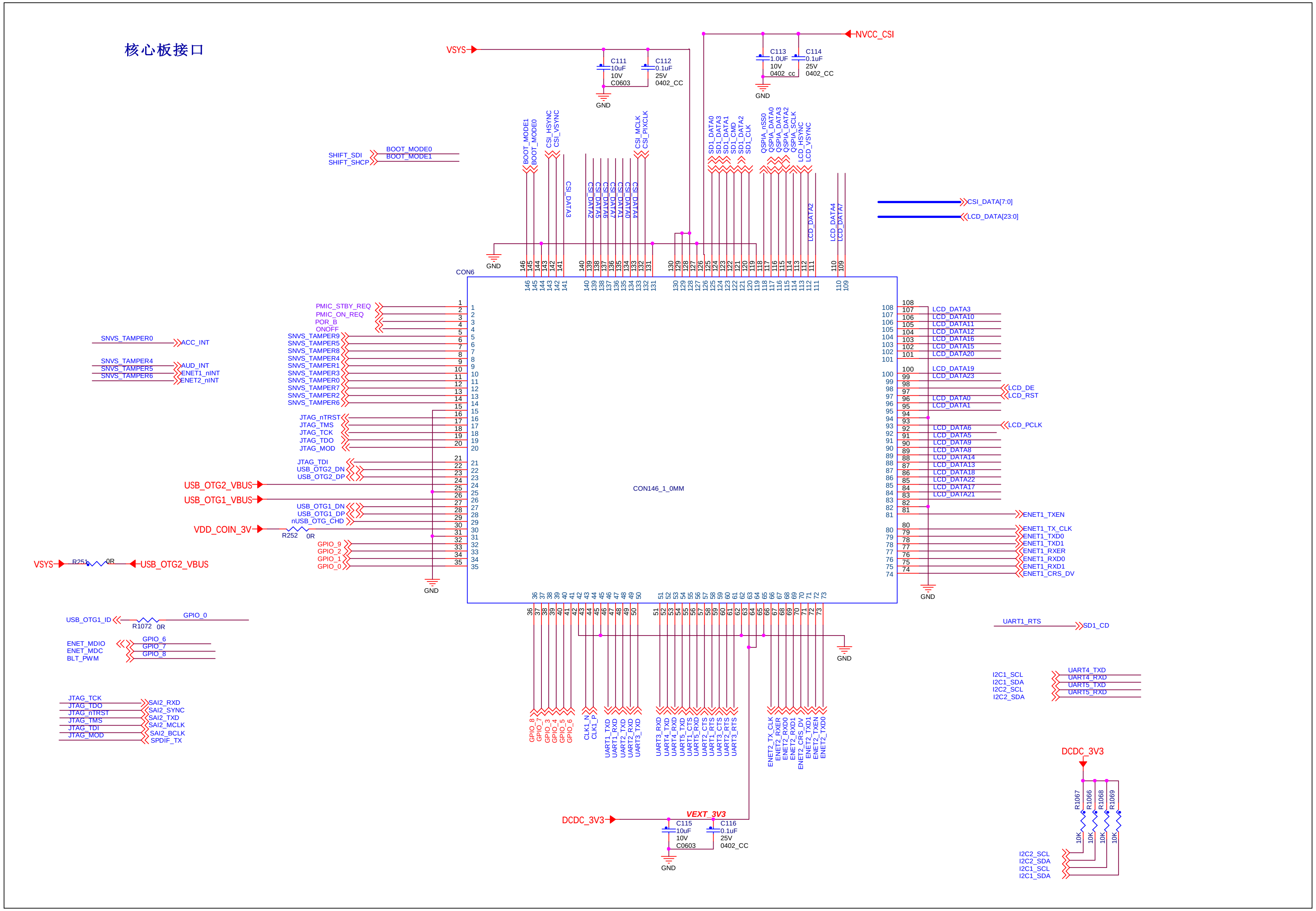
Size	Document Number	Rev
Custom	<Doc>	1.0

Date: Tuesday, March 24, 2020 Sheet 12 of 14

HDMI原理



Title	HDMI	
Size C	Document Number <Doc>	Rev 1.0
Date: Tuesday, March 24, 2020	Sheet 13 of 14	



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Title

BOARD CONNECTOR

Size	Document Number	Rev
C		V1.0

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